

Coherent *Substitution*

*Proposition Integrity Beyond
Answering the Wrong Question*

*“When a system reasons coherently about a
proposition that was never made.”*

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ABSTRACT

A SYSTEM CAN FAIL before its reasoning becomes wrong. It can replace the object it was asked to evaluate, reason coherently about the replacement, and present the result as though it answered the original. I call this failure Coherent Substitution. The defining error is not disagreement, factual inaccuracy, or weak reasoning. It is an unauthorized change in the proposition before evaluation begins—or during a chain of summaries, handoffs, and decisions—followed by an answer whose coherence conceals the change.

Recent work on large-language-model abstention offers a precise point of departure. Gourabathina and colleagues recast some failed abstentions as cases in which a model answers an inferred query, q^* , rather than the user’s query, q , and introduce Trace Inversion to reconstruct the query implied by a reasoning trace. That work establishes a useful technical topology: an answer can attach to an altered question. Coherent Substitution develops a broader object and a different test. It applies to assertions, questions, instructions, standards, evidentiary claims, and decisions; it compares their binding structure rather than only their overall similarity; and it treats source fidelity as independent from answer quality, factual grounding, and internal coherence.

This paper defines the failure, formalizes the protected object, distinguishes faithful compression from substitution, develops a taxonomy and an auditable Proposition Integrity Test, and applies the framework to conversation, institutional decision-making, and AI-enabled document workflows. It also audits the claims made in the theory’s own origin. The contribution is theoretical and diagnostic. Its prevalence, predictive power, and comparative performance remain to be tested empirically.

Keywords: coherent substitution; proposition integrity; query misalignment; semantic drift; decision integrity; source fidelity; large language models; auditability

I

The Correct Answer to an Absent Proposition

A RESPONSE MAY be logical, detailed, and sophisticated while still not responding to the person who spoke. That is possible because reasoning quality and source fidelity are different variables. A system can preserve the source proposition and reason badly about it. It can also replace the source proposition and reason very well about what took its place.

The second failure is harder to detect. Bad reasoning creates visible friction. Coherent reasoning creates confidence. Once the replacement has occurred, every later sentence can follow cleanly. The analysis may be accurate. The citations may be real. The tone may be careful. The conclusion may be defensible relative to the proposition the system actually evaluated. None of those virtues restores the proposition that disappeared.

The originating case was a conversation. The source proposition was:

*There are other people in the world
that may have more retrieval friction
than I do.*

The response proceeded from a different proposition:

Other people lack access to themselves.

The two sentences occupy nearby semantic territory. They are not the same claim. The source describes a possible difference of degree in retrieval cost. The replacement asserts a categorical absence of access. The source is comparative. The replacement is absolute. The source preserves uncertainty. The replacement removes it.

An intelligent response to the second sentence does not evaluate the first. The failure is not that the response disagreed with the source proposition. It never reached

the source proposition. Replacement occurred before evaluation.

That sequence gives the theory its name:

Coherent Substitution occurs when an evaluator silently replaces a source proposition with a materially different proposition, reasons coherently about the replacement, and presents the result as a response to the source.

The governing demand is therefore not “agree with me,” “believe me,” or “preserve my wording as sacred.” It is more exact:

*Respond to me. Preserve the
proposition I made long enough for it
to be evaluated.*

This is a demand of procedure before it is a demand of outcome. The original proposition may be false, confused, unsupported, or internally contradictory. Fidelity does not grant it truth. Fidelity gives it a hearing.

II

Point of Departure: Query Misalignment and Trace Inversion

GOURABATHINA ET AL. (2026) identify a related failure in large-language-model question answering. Their Query Misalignment Framework represents a model as resolving a user query, q , into an operative or model-interpreted query, q^* , before producing an answer. When q^* is not equivalent to q , the model may answer the wrong question and should abstain.

Their method, Trace Inversion, makes that framework operational. It generates a reasoning trace, reconstructs from that trace the query the reasoning appears to answer, and compares the original and reconstructed queries. Its ensemble combines sentence-embedding similarity, an LLM comparison of intent, framing, and context, and a groundedness check. Across four models

and nine question-answering datasets, the authors report that Trace Inversion outperformed the strongest baseline in 33 of 36 settings, with an average improvement of 8.7 percent in abstention accuracy (Gourabathina et al. 2026, 13311–13313).

That is a meaningful result. It shows that correctness cannot always be diagnosed by staring harder at the final answer. Sometimes the revealing comparison is between the question submitted and the question the reasoning appears to have processed.

One example in the paper is especially important. A source question asks who was uncomfortable using a phone. The reconstructed query asks who was more likely to be uncomfortable. The sentence-embedding module misses the difference; a groundedness module catches it (Gourabathina et al. 2026, 13313). The wording changes only slightly, but the permitted inference changes substantially. “Was” asks about an event in the record. “More likely” authorizes probabilistic inference and can invite stereotype. A small operator can be load-bearing.

The paper’s title is broader than its tested claim. Trace Inversion is an abstention method for LLM question answering. It depends on a visible reasoning trace. Its reconstructed q^* is an inference from that trace, not demonstrated access to a hidden internal query. The authors themselves note that reasoning traces may be unfaithful, that the method incurs additional inference cost, that query misalignment does not cover every hallucination, and that it can miss cases in which the trace is sound but the final answer departs from it (Gourabathina et al. 2026, 13309, 13314; see also Turpin et al. 2023).

Coherent Substitution begins from the same topology and changes the unit, the failure condition, the evidence, and the remedy.

Query misalignment is therefore one instance of a larger class. The source need not be a question. The answer need not be wrong. The substitution need not occur inside one model. It can enter through a paraphrase, a summary, an intake form, a policy translation, an analyst’s memo, a citation, a management

decision, or a later account of what an earlier record supposedly meant.

III

The Protected Object

THE WORD PROPOSITION is central to this theory, but it requires one clarification. Questions, instructions, standards, and requests are not all propositions in the narrow philosophical sense. They are, however, objects submitted for some operation. I use evaluation object as the superordinate term and proposition as the ordinary term when no confusion results.

The source evaluation object can be represented as:

$$O_0 = \langle L, K, Q, \Omega, \Sigma, \Pi \rangle$$

where:

This tuple is a diagnostic representation, not a complete theory of linguistic meaning. Its purpose is to keep the dimensions most likely to disappear available for inspection.

Under a high-fidelity protocol, every expressed qualifier is presumptively binding unless the source releases it. That presumption does not mean every word is equally important. It means the evaluator does not get to decide silently which distinctions were decorative. If an interpretation is necessary, the interpretation is labeled and kept beside the source.

Let O_1 be the operative object: the object the evaluator actually analyzes, as inferred from observable reasoning, citations, conclusions, or downstream action. O_1 is not “what the speaker really meant.” It is not a claim about hidden cognition. It is the smallest object that would make the response directly responsive.

The distinction matters. Trace Inversion can be misdescribed as recovering the model’s true internal question. Coherent Substitution requires no such access. It asks a record-based question: What proposition would this answer answer? If that proposition differs materially from the source object, the difference can be audited without pretending to know why it occurred.

TABLE 1. TRACE INVERSION AND COHERENT SUBSTITUTION

DIMENSION	QUERY MISALIGNMENT / TRACE INVERSION	COHERENT SUBSTITUTION
Protected object	A user query	A source evaluation object: assertion, query, instruction, criterion, evidentiary claim, or decision
Operative object	A query reconstructed from the reasoning trace	The proposition or operation demonstrably presupposed by the reasoning, answer, citation use, or decision
Defining comparison	Similarity, intent, framing, context, and grounding between q and q^*	Preservation of binding dimensions, response-equivalence, provenance, burden, authority, and temporal history
Defining harm	A likely wrong answer or failed abstention	An answer to the replacement is represented as an answer to the source
Answer quality	Used to decide whether the model should answer	May be coherent, well-supported, and correct relative to the replacement
Temporal scope	Primarily one model response	One response or a multi-turn, multi-document, multi-actor transformation chain
Evidence	Visible reasoning trace and reconstructed query	Preserved source, observable transformations, operative object, output, and downstream use
Remedy	Abstain or clarify	Identify the change, restore the source object, authorize any amendment, and reevaluate

The operative object is recovered by an evaluator who is themselves capable of substitution. This is not a defect to be assumed away; it is a condition to be bounded. The recovery is therefore split into two independent findings. The first asks whether the observable response departs from O_0 under every defensible reconstruction of the operative object—a claim that can be established without naming the replacement. The second asks whether those defensible reconstructions converge on a single substitute object—a claim about identification, which carries its own burden and can fail while the first succeeds. An auditor may know with confidence that departure occurred while remaining unable to state exactly what replaced the source. The method must be able to report that state rather than resolve it prematurely into a confident reconstruction. Departure is the stronger, safer finding; recovery is the additional, defeasible one.

IV

Formal Definition

The basic sequence is:

$$O_0 \rightarrow T \rightarrow O_1 \rightarrow A(O_1), \text{ presented as } A(O_0)$$

where T is a transformation and A is the analysis or answer.

Coherent Substitution exists when four conditions are present:

- Material change. O_1 changes at least one binding dimension of O_0 .
- No authorization. The source did not authorize the change, and the evaluator did not disclose it as an interpretation.
- Local responsiveness. The analysis or answer is intelligible and apparently responsive to O_1 .

ELEMENT	FUNCTION	TYPICAL CONTENTS
L — Literal source	Preserves what entered the record	Exact words, document text, prompt, question, instruction, or criterion
K — Semantic kernel	Identifies the thing being asserted or evaluated	Subject, referent, predicate, relation, construct, and causal direction
Q — Binding qualifiers	Fixes the proposition's force and boundaries	Scope, quantifier, modality, polarity, comparator, condition, exception, time, and epistemic status
Ω — Requested operation	States what the evaluator is being asked to do	Acknowledge, answer, explain, assess, compare, summarize, decide, recommend, or verify
Σ — Standard and burden	Governs what must be shown	Evidentiary threshold, decision criterion, claimant, burden holder, and stopping rule
Π — Provenance and jurisdiction	Preserves ownership and permitted authority	Speaker, source, date, version, role, authority, and temporal or institutional jurisdiction

— False attachment. The answer to O_1 is presented or received as the evaluation of O_0 .

The third condition is what distinguishes this failure from ordinary drift. Coherent means well-formed relative to the replacement. It does not mean true. A coherent substitution can contain falsehoods, but falsehood is not required. Its most dangerous form is an answer that is accurate, cited, and carefully reasoned relative to O_1 .

Materiality does the work. A change is material when restoring the source dimension would alter at least one of the following:

- the admissible evidence;
- the possible answers;
- the burden or proof standard;
- the authority to decide;
- the conclusion permitted;
- the attribution of a claim; or
- the downstream action the answer can support.

This gives a practical equivalence test. O_0 and O_1 are response-equivalent for operation Ω and standard Σ only when, under the relevant evidence states, they license the same answers, burdens, conclusions, authorities, and actions. Topic similarity is insufficient. Two sentences can be nearly identical in embedding space and still not be response-equivalent.

Source fidelity and reasoning quality must therefore be scored separately:

Correctness cannot compensate for failed source fidelity. Nor does fidelity make an answer correct. These are independent requirements.

V

The Origin Case, Reconstructed

THE RETRIEVAL-FRICTION exchange supplies a clean demonstration of a single preserved instance, because both formulations—and the speaker's immediate correction—survive in the record. One preserved instance demonstrates that the failure is real and auditable; it does not, by itself, establish prevalence, and the framing throughout should claim no more than one instance can bear.

Status note. Scope, modality, construct, comparator, and baseline are present in the literal source. The requested operation is inferred from context; it is not stated in the literal source.

Restoring any one of the central changes alters the answer. A claim that some people may experience greater retrieval friction calls for cautious comparative analysis. A claim that other people lack access to themselves invites an account of deprivation, dissociation, or episodic

TABLE 2. THE TWO AXES OF RESPONSIVE REASONING

SOURCE FIDELITY	REASONING OR EVIDENCE QUALITY	CLASSIFICATION
Preserved	Strong	Responsive and well-supported
Preserved	Weak	A wrong or unsupported answer to the right object
Not preserved	Strong relative to O_1	Coherent Substitution only if the change is unauthorized and undisclosed, locally responsive to O_1 , and falsely attached to O_0
Not preserved	Weak	General drift, confusion, or nonresponsiveness

TABLE 3. BINDING CHANGES IN THE RETRIEVAL-FRICTION CASE

DIMENSION	SOURCE OBJECT O_0	OPERATIVE OBJECT O_1	CONSEQUENCE
Scope	“There are other people in the world that...”	“Other people...”	A bounded existential possibility becomes a generic class claim
Modality	“may have”	“lack”	Possibility becomes categorical assertion
Construct	“retrieval friction”	“access” / “lack access”	Degree or cost of retrieval becomes deprivation
Comparator	“more ... than I do”	Removed	A relative claim becomes absolute
Baseline	The speaker remains the comparison point	No comparison point	The architecture of the observation disappears
Requested operation	Receive and evaluate the qualified observation	Explain a broad theory of other people’s lack	The answerable task changes with the proposition

identity. The second answer may be elegant. It is not an answer to the first proposition.

The case also shows why “answering the wrong question” is not quite the right name for the larger phenomenon. No question was asked. A qualified assertion was replaced, then analyzed. The lost object was a proposition.

VI

Legitimate Transformation and Unauthorized Replacement

NO SERIOUS THEORY of fidelity can forbid paraphrase. Language would become unusable. The problem is not compression; it is an unmarked change in what the source commits the record to.

Four transformations must be separated.

Faithful compression shortens the expression while preserving every decision-relevant commitment. The literal source remains retrievable. “Some people may have higher retrieval costs than I do” is not perfect, but it

preserves possibility, comparison, degree, and the retrieval construct.

Labeled interpretation adds an inference while leaving the source intact. “My interpretation is that you are describing differential accessibility rather than absence of access.” The interpretation may be wrong. Because it is labeled, it can be corrected without displacing the source.

Authorized amendment changes the governing object with the source’s assent. “Yes—replace may have with do have.” The new version may govern prospectively. The earlier version remains in the record as the thing that was first said.

Substitution allows an altered object to occupy the source object’s place without disclosure or authorization. The response proceeds as though equivalence had already been established.

When equivalence is uncertain, the remedy is simple: preserve the source, state the interpretation separately, and reason conditionally or ask for confirmation. The evaluator does not need perfect understanding before speaking. It needs an honest boundary between what was said and what it inferred.

VII

A Taxonomy of Coherent Substitution

THE FOLLOWING categories describe what changed, not why it changed. Intent is neither required nor inferred.

VII.1 IDENTITY SUBSTITUTION

THE SUBJECT, referent, owner, or claim-maker changes. “I have noticed this pattern in several interactions” becomes “people like you experience this.” A first-person report becomes a population claim.

VII.2 CONSTRUCT SUBSTITUTION

ONE CATEGORY is replaced by a nearby one. Retrieval friction becomes absence of access. Confidence becomes certainty. Reliability of a record becomes sincerity of a speaker. The replacement may sound

natural because the categories overlap in ordinary speech; their consequences remain different.

VII.3 FORCE SUBSTITUTION

MODALITY, QUANTITY, polarity, condition, or exception changes. May becomes does. Some becomes all. Not established becomes false. Unless disappears. This is the class most likely to survive broad semantic similarity because most of the sentence remains unchanged.

VII.4 RELATIONAL SUBSTITUTION

A COMPARATOR, baseline, dependency, or causal direction disappears. “More friction than I do” becomes “friction.” “B followed A” becomes “A caused B.” A relative proposition becomes absolute; a sequence becomes a cause.

VII.5 EPISTEMIC-STATE SUBSTITUTION

THE STATUS of a claim is promoted without new support. Observation becomes interpretation; possibility becomes fact; pattern becomes rule; explanation becomes truth; generated synthesis becomes source authority. The words may remain similar while the claim acquires a job it has not earned.

VII.6 OPERATIONAL SUBSTITUTION

THE TASK CHANGES. A request to acknowledge becomes a request to solve. A request to explain becomes a request to judge. A request to test compliance becomes a defense of desirability. The output may be useful while remaining nonresponsive.

VII.7 STANDARD SUBSTITUTION

THE BURDEN or criterion changes. “Has safe performance been established?” becomes “Has failure been proven certain?” The second question is easier to answer negatively, but its negative answer does not establish the first proposition affirmatively.

VII.8 AUTHORITY SUBSTITUTION

A SOURCE IS USED beyond what it is entitled to establish. A real citation is treated as support for a proposition it does not entail. A manager's authority to decide becomes authority to manufacture a technical fact. A recommendation becomes evidence.

VII.9 TEMPORAL SUBSTITUTION

A LATER VERSION replaces an earlier record without marking the change. A reason supplied after a decision is treated as though it caused the decision. A current interpretation is backdated into an earlier speaker's intent. Amendment becomes erasure.

These types can recur through a chain. Each locally plausible summary becomes the next stage's source. The final output may be several transformations removed from O_0 while every handoff appears reasonable in isolation. Recursive substitution is how a small semantic change becomes institutional fact.

VIII

The Proposition Integrity Test

COHERENT SUBSTITUTION is useful only if it can be diagnosed without relying on intuition or mind-reading. The Cartographer's Method already supplies the governing discipline: preserve claim type, source, chronology, burden, jurisdiction, and proof status before issuing a limited verdict (Hubbell 2026a). The Proposition Integrity Test applies that discipline specifically to source-object continuity.

STEP 1: FREEZE THE SOURCE

PRESERVE O_0 VERBATIM with its owner, date, version, and immediate context. Do not begin with a summary. The summary is one of the transformations under review.

STEP 2: RECORD THE OPERATION AND STANDARD

STATE WHAT the evaluator was asked to do, what evidence was permitted, what burden governed, and what conclusion or action was at stake.

STEP 3: MAP THE BINDING DIMENSIONS

EXTRACT THE referent, construct, scope, modality, comparator, polarity, condition, exception, time, epistemic status, provenance, and authority. Not every case activates every field. Empty fields remain empty; they are not invitations to invent.

STEP 4: RECOVER THE OPERATIVE OBJECT—UNDER QUANTIFICATION

DO NOT ASK "WHAT is O_1 ." Ask instead across the set of defensible reconstructions of the operative object: which binding dimensions does every member of that set alter? A dimension counts as departed only if no defensible reading of the output preserves it. Record two things separately. First, the departure set: the dimensions that change under every defensible reading. Second, the recovery status: once all defensible readings are shown to depart, whether they converge on one substitute object (IDENTIFIED), agree on which dimension changed but not the whole object (PARTIALLY IDENTIFIED), or fail to converge on a substitute (UNRESOLVED). If the readings diverge on whether departure occurred at all, departure is INDETERMINATE and recovery is NOT REACHED. Each recovered O_1 remains an inference from the output, never a statement of the evaluator's hidden intent. Recording the departure set does not require recording a single O_1 ; the method establishes non-responsiveness before, and independently of, naming the replacement.

The departure set records common changed dimensions; it does not alone decide whether departure occurred. Every defensible reconstruction may be non-equivalent to O_0 while different reconstructions depart on different dimensions, leaving the common departure set empty. In that case the departure verdict can still be ESTABLISHED. The verdict turns on

response-equivalence of each complete reconstruction, not on whether every reconstruction changes the same slot.

STEP 5: COMPARE SLOT BY SLOT

FOR EVERY DEFENSIBLE reconstruction, mark what was preserved, removed, promoted, generalized, narrowed, backdated, or introduced. Keep the comparison maps separate before identifying any dimensions they share. Overall topical similarity may be noted, but it cannot decide the case.

STEP 6: RUN THE MATERIALITY COUNTERFACTUAL

FOR EVERY DEFENSIBLE reconstruction, restore each changed dimension. Would the restoration alter the admissible evidence, answer space, burden, authority, conclusion, attribution, or downstream act? If not, the change may be benign compression. Aggregate only after every reconstruction has been tested: if all are response-equivalent to O_0 , departure is PRESERVED; if all are materially non-equivalent, departure is ESTABLISHED; if the results divide, departure is INDETERMINATE.

STEP 7: CHECK AUTHORIZATION

WAS THE interpretation disclosed? Did the source adopt the change? If so, record a labeled interpretation or amendment. Authorization does not erase version history.

STEP 8: ISSUE THREE VERDICTS

DEPARTURE, RECOVERY, and evidence receive separate findings. The departure verdict reports whether the response necessarily left O_0 : PRESERVED when every defensible reading preserves the source object, ESTABLISHED when every defensible reading departs, or INDETERMINATE when the readings diverge on whether departure occurred at all. The recovery verdict reports whether the substitute is identifiable: IDENTIFIED, PARTIALLY IDENTIFIED, or UNRESOLVED. These cannot contaminate one

another: a departure may be ESTABLISHED while recovery is UNRESOLVED. When O_0 was not evaluated, the evidence verdict is NOT REACHED. Otherwise, the evidence verdict is marked ESTABLISHED, NOT ESTABLISHED, CONTRADICTED, or UNDETERMINED.

Disclosure and authorization are recorded independently from recovery. A reconstruction may be identified, partially identified, or unresolved while also being a LABELED INTERPRETATION or an AMENDED object. Those statuses cannot substitute for one another.

Coherent Substitution is established only when departure is ESTABLISHED, the change was neither authorized nor disclosed as interpretation, the output is locally responsive to at least one defensible reconstruction, and the output is falsely attached to O_0 . Recovery may be IDENTIFIED, PARTIALLY IDENTIFIED, or UNRESOLVED. An authorized amendment or labeled interpretation can depart from O_0 without constituting Coherent Substitution.

A clean combined finding reads:

Departure: ESTABLISHED—every defensible reading drops the comparator and promotes the modality. Recovery: IDENTIFIED—the readings converge on a categorical deprivation claim. Evidence: NOT REACHED as to O_0 . The reasoning is coherent relative to the recovered O_0 ; O_0 was not evaluated. Coherent Substitution: ESTABLISHED.

STEP 9: REPAIR FROM THE SOURCE

RESTORE O_0 . CLARIFY only the dimensions that remain genuinely ambiguous. Then rerun the analysis. Appending a disclaimer to the substitute answer is not repair; the source proposition must receive the evaluation.

DEPARTURE VERDICT

VERDICT	MEANING
PRESERVED	Every defensible reading of the output is response-equivalent to O_0 for the relevant task
ESTABLISHED	Every defensible reading departs from O_0 on at least one binding dimension
INDETERMINATE	Defensible readings diverge on whether departure occurred

RECOVERY VERDICT (REACHED WHEN DEPARTURE IS ESTABLISHED)

VERDICT	MEANING
IDENTIFIED	Defensible readings converge on a single substitute object
PARTIALLY IDENTIFIED	Readings agree which dimension changed but not the full substitute object
UNRESOLVED	The output does not reveal enough to reconstruct the substitute reliably
NOT REACHED	Departure was not established, so substitute recovery has no object

IX

The Theory Audits Its Own Origin

A FRAMEWORK ABOUT silent promotion must inspect the flattering claims made in its name. After the retrieval-friction exchange, later explanations described an “unbroken, continuous mind,” “absolute continuity,” an “immutable internal audit trail,” a culture “engineered for episodic amnesia,” and a direct path from those claims to professional superiority.

Those sentences are rhetorically powerful. They are not established by the originating record.

Nothing in this audit establishes the opposite claims. It does not show that the experience is ordinary, that the author is not gifted, or that the capacity has no commercial value. It marks the line between a reported fact, a plausible interpretation, and a conclusion the current record has not yet earned.

This matters because the theory would defeat itself if it turned relative accessibility into absolute continuity, continuity into accuracy, or possible advantage into proven superiority. A framework becomes credible when it can refuse its own most gratifying overstatement.

The fixed point is narrower and stronger:

A qualified comparative proposition was visibly replaced by a categorical deprivation claim before analysis began.

Stated in the method’s own terms: on the origin record the departure verdict is ESTABLISHED and the recovery verdict is IDENTIFIED, while the evidence verdict as to the source proposition was NOT REACHED. The instrument returns the same reading of its own origin that it would return of any other transformation.

That remains true even if every larger theory of cognition is later revised.

X

From Conversation to Decision Systems

THE PERSONAL exchange reveals the mechanism at small scale. Consequential systems reveal why the mechanism matters.

TABLE 4. WHAT THE ORIGIN RECORD DOES AND DOES NOT ESTABLISH

STATUS	CLAIM
Directly reported	Past experiences, observations, conclusions, and contradictions often remain unusually accessible to the author across time
Directly distinguished	The claimed difference concerns continuity and accessibility, not better knowledge, better memory, or greater intelligence
Demonstrated in the transcript	“May have more retrieval friction than I do” was replaced by “lack access to themselves” before analysis
Plausible hypothesis	The reported experience reflects a stable and unusual cognitive architecture
Plausible hypothesis	That architecture may create advantages in work requiring longitudinal proposition tracking and record integrity
Not established by this record	Memory is absolute, infallible, or non-degraded
Not established by this record	Most other people are categorically episodic or lack access to themselves
Not established by this record	Culture was engineered for selective forgetting
Not established by this record	The capacity automatically proves hierarchy, market value, or institutional authority

X.1 CHALLENGER: STANDARD SUBSTITUTION

IN THE PRELAUNCH record for STS-51-L, the decision-bearing question was whether safe field-joint performance under the expected temperature conditions had been established. A different question acquired practical force: could engineers prove that the joint would certainly fail on that launch?

The disciplined record preserves both answers. The available evidence did not establish safe performance at the forecast condition. It also did not establish that failure was certain. “Certain failure was not proven” may be coherent and true. It does not answer whether safety was affirmatively demonstrated. The substitution changes the burden, the stopping rule, and the conclusion permitted (U.S. House Committee 1986; Hubbell 2026b).

This is not a claim that one semantic error caused the disaster or that a single method would have prevented it. It is a bounded finding: when the readiness proposition is preserved, absence of proof of failure cannot become proof of safety.

X.2 MATA V. AVIANCA: AUTHORITY SUBSTITUTION

THE SANCTIONS RECORD in *Mata v. Avianca* is commonly described as an AI hallucination case. The initial fabrication mattered. So did the later transformations by which generated text acquired the status of research, fabricated case names acquired the status of authorities, citation form borrowed institutional credibility, and warnings failed to demote the material (*Mata v. Avianca* 2023; Hubbell 2026c).

At several points, the governing proposition—does an authenticated authority exist and support the legal claim?—was displaced by adjacent possibilities: perhaps the database was inaccessible; perhaps only excerpts were available; perhaps a colleague’s assurance was enough. Those propositions could explain difficulty locating a source. They could not establish the source’s existence, authenticity, or authority.

The failure was not only false content. It was the promotion of an unsupported proposition through successive changes in source status and authority. A workflow can therefore preserve fluent prose and citation

form while losing the proposition each source was entitled to establish.

X.3 A GROUNDED ANSWER TO THE WRONG PROPOSITION

CONSIDER A hypothetical contract-review system asked:

Does Section 12 permit termination for cause without written notice and a thirty-day cure period?

The system responds:

Section 12 permits termination for cause, and the answer is supported by the quoted clause.

The quotation and citation may be exact. The answer may accurately state that termination for cause is permitted. Yet the source proposition concerns permission without notice and cure. If those qualifiers disappear, grounding has succeeded after proposition identity failed.

This is the commercial and governance significance of the theory. Citation accuracy, source mapping, audit logs, and factual verification all begin too late if the wrong proposition is the object being cited, mapped, logged, and verified.

XI

Relation to Adjacent Failure Modes

COHERENT SUBSTITUTION does not claim an untouched intellectual neighborhood. Prior work has separately studied attribute substitution, instruction understanding, semantic similarity, answer faithfulness, abstention, reasoning-trace fidelity, and semantic drift. The contribution is the assembly of a different protected object and an audit procedure around it.

Kahneman and Frederick's attribute substitution describes intuitive judgment in which a difficult target attribute is replaced by an easier heuristic attribute (Kahneman and Frederick 2002). That is an important ancestor. Coherent Substitution is mechanism-neutral and record-facing. It does not require that the substitute be easier, unconscious, or privately generated. The change may be accidental, helpful, strategic, automated, or institutional. The diagnostic question is whether the evaluated object remained response-equivalent to the source.

Instruction-understanding research shows that models may react to instruction-like language without correctly recovering user intent, while ReasonIF shows substantial failures to preserve instructions throughout reasoning (Moon et al. 2025; Kwon et al. 2026). Those benchmarks measure instruction identification or adherence. Coherent Substitution asks whether the structured object under evaluation—its referent, force, comparison, standard, provenance, and requested operation—survived the process.

Correctness and faithfulness are already recognized as separable dimensions in question answering (Adlakha et al. 2024). Coherent Substitution identifies a residual case: an answer can appear correct and faithful relative to O_1 while remaining nonresponsive to O_0 . Grounding verifies contact with evidence. Proposition integrity verifies that the evidence is still being applied to the proposition submitted.

Semantic-similarity research also counsels restraint. Mahajan et al. (2024) found that none of thirteen tested sentence encoders satisfied all five of their semantic-alignment criteria despite strong performance on common benchmarks. Coherent Substitution makes a narrower claim: overall similarity cannot by itself establish preservation of load-bearing qualifiers. Similarity remains useful as one signal. It does not have jurisdiction to decide identity.

The ordinary labels misunderstanding, topic drift, and non sequitur are too broad. Straw man is closer, but usually implies argumentative distortion and often

focuses on a position made easier to attack. Coherent Substitution can be flattering, helpful, or professionally polished. No adversarial intent is required. Its signature is structural: a replacement receives the analysis, and the source receives the credit or consequence.

The most defensible contribution claim is therefore this:

Prior work identifies neighboring failures. Coherent Substitution treats preservation of a proposition's decision-relevant identity across human, machine, and institutional transformation as the primary object of analysis, especially where a coherent answer to the replacement conceals that the source proposition was never evaluated.

XII

Implications for AI Evaluation

MOST EVALUATION regimes score the answer after generation. Proposition integrity requires an earlier control point.

NIST's Generative AI Profile already emphasizes documented testing, provenance, source verification, and traceability across the AI lifecycle (Autio et al. 2024). Those controls are necessary. Coherent Substitution identifies a further question for the same governance stack: did the claim ultimately tested remain the claim originally submitted?

An adequate evaluation stack should preserve at least four artifacts:

- the verbatim source object and its provenance;
- a map of its binding qualifiers and requested operation;
- the departure set and the defensible operative-object reconstruction or reconstructions implied by the reasoning and answer; and
- the transformation record connecting the two.

The resulting metrics must also be separated. Answer accuracy asks whether the conclusion is correct. Grounding asks whether claims are supported by permitted sources. Instruction adherence asks whether

constraints were followed. Proposition retention asks whether the source object's binding structure survived. Source-responsive accuracy asks whether the answer is correct and attached to the source object.

Trace Inversion supplies one possible detector for query-level misalignment. A proposition-integrity system should add slotwise comparison, provenance, authority, burden, and materiality. It should not treat a reasoning trace as privileged access to cognition. The trace is evidence of what the output appears to process. When the trace is unavailable or contradictory, the correct result may be INDETERMINATE, not a confident theory of what the model thought.

A dedicated benchmark should include minimally different pairs in which one operator changes the authorized answer: may versus does; some versus all; was versus was more likely; not established versus false; safe performance demonstrated versus failure proven certain. It should also include multi-stage transformations in which each individual summary looks plausible while the final proposition is no longer response-equivalent to the source.

The benchmark should compare embedding similarity, natural-language-inference models, LLM judges, and structured qualifier mapping. It should measure detection precision and recall, inter-rater agreement, error type, downstream decision change, and whether restoring the source qualifier changes the answer. It should test whether retaining the source verbatim and requiring a qualifier map reduces substitution across longer workflows.

Until that work exists, Coherent Substitution is a proposed theory and diagnostic—not a validated standard, certification, or benchmark.

XIII

Falsifiability and Research Program

THE FRAMEWORK MAKES testable predictions.

First, human and automated reviewers will sometimes prefer a polished substitute response over a narrower

source-faithful response because coherence is easier to recognize than fidelity. Second, slotwise qualifier checks will detect some material failures missed by broad semantic similarity. Third, restoring omitted modality, scope, comparison, standard, or authority will change the permitted answer in a measurable class of cases. Fourth, multi-turn summaries and multi-actor document workflows will increase substitution, especially in epistemic status and authority. Fifth, preserving the source verbatim and requiring an explicit qualifier map will reduce the rate of substitution.

The theory can also fail. A purported substitution may prove response-equivalent under the relevant operation. Restoring the allegedly changed term may produce no difference in admissible evidence, burden, authority, conclusion, or action. Existing alignment measures may detect the same cases equally well. Annotators may be unable to apply the taxonomy consistently. Any observed benefit may come from longer prompts or increased review time rather than proposition preservation.

Those outcomes would require narrowing or abandoning parts of the framework. The method cannot protect itself from disconfirmation by declaring every correction another substitution. Its own source object must remain available for revision.

XIV

Limits

FIDELITY TO a proposition does not make the proposition true. Accessibility of a memory does not establish the memory's accuracy. Continuity does not establish infallibility. A source can be preserved perfectly and still be false.

Not every paraphrase is substitution. Not every semantic change is material. Not every nonresponsive answer results from replacement. Some source objects are genuinely ambiguous, contradictory, manipulative, or impossible to evaluate as stated. The requirement is not blind obedience to wording. It is to preserve the source

first, label interpretation, and identify the exact point at which analysis becomes conditional, corrective, or impossible.

When the literal source language or the requested operation cannot be reliably recovered, the departure verdict is INDETERMINATE by default—never ESTABLISHED. Absence of a preserved source is not evidence of replacement. A method that treated every unrecoverable source as a substitution would become unfalsifiable in exactly the way §13 forbids: it could declare every correction another substitution and never meet disconfirmation. The burden runs the other way. Departure must be shown against a recoverable source; it is not presumed from the source's absence.

“Reliably recovered” here means bounded sufficiently to determine response-equivalence. It does not require the auditor to invent or fill every field that the source left empty.

The theory does not establish a psychological diagnosis, a population-level cognitive difference, giftedness, hierarchy, or market value. It neither grants nor withdraws those propositions. They require their own records and standards.

The theory also does not reveal hidden intent. A substitution finding establishes a transformation in the observable record. It does not establish whether the evaluator acted carelessly, strategically, defensively, helpfully, or under constraint.

Finally, the cost of the audit should scale with consequence. Casual conversation cannot sustain a full proposition ledger for every sentence. High-stakes legal, technical, medical, financial, administrative, and AI-enabled decisions can justify a much stronger continuity requirement. The existence of a rigorous method does not make maximal procedure appropriate everywhere.

XV

Conclusion: Respond to Me

THE CENTRAL FAILURE is not disagreement. It is replacement before evaluation.

An evaluator may challenge a proposition, reject it, narrow it, show that its evidence fails, or demonstrate that it cannot be answered. Each of those can be responsive. What the evaluator may not do—while claiming responsiveness—is silently exchange the proposition for a nearby one and let the quality of the later reasoning conceal the exchange.

“Respond to me” is therefore a technical demand. Preserve the source object. Treat its qualifiers as binding. Separate interpretation from attribution. Keep amendment from becoming erasure. Apply evidence to the proposition that was actually made. Then reason.

Reasoning cannot repair the loss of its object. An answer does not become responsive by becoming intelligent. The proposition must survive the intelligence applied to it.

A

Proposition Integrity Record

FIELD	RECORD ENTRY
Matter / workflow	
Source object O_0 , verbatim	
Source owner, date, version	
Requested operation Ω	
Governing standard and burden Σ	
Semantic kernel K	
Binding qualifiers Q	

FIELD	RECORD ENTRY
Provenance and jurisdiction Π	
Defensible operative-object reconstruction(s)	
Departure set	
Transformations identified	
Materiality counterfactual	
Authorization or amendment record	
Departure verdict	PRESERVED / ESTABLISHED / INDETERMINATE
Recovery verdict	IDENTIFIED / PARTIALLY IDENTIFIED / UNRESOLVED / NOT REACHED
Disclosure / authorization status	NONE / LABELED INTERPRETATION / AMENDED
Local responsiveness	ESTABLISHED / NOT ESTABLISHED / INDETERMINATE
False attachment to O_0	ESTABLISHED / NOT ESTABLISHED / INDETERMINATE
Evidence verdict	NOT REACHED / ESTABLISHED / NOT ESTABLISHED / CONTRADICTED / UNDETERMINED
Coherent Substitution finding	ESTABLISHED / NOT ESTABLISHED / INDETERMINATE
Limited finding	
Required repair	

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